

Juan David Muñoz-Bolaños

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SmarAct Metrology GmbH & Co. KG

z. Hd. Melissa Skrivanek

Oldenburg

Deutschland

Bewerbung als Applikationsingenieur (m/w/d), Laserinterferometrie und Vibrometrie

Sehr geehrte Frau Skrivanek,

hiermit bewerbe ich mich als Applikationsingenieur im Bereich Laserinterferometrie und Vibrometrie bei SmarAct Metrology. Ich bin ein ursprünglich aus Kolumbien stammender Physikingenieur und promoviere derzeit an der Medizinischen Universität Innsbruck. In meine künftige Rolle bringe ich fundierte Erfahrung in interferometrischer Messtechnik, Prototypenentwicklung und Datenauswertung ein.

Kern meiner Doktorarbeit ist ein interferometrisches Wellenfrontsensormesssystem, das ich eigenständig aufgebaut habe. Der optische Aufbau basiert auf einem Michelson-Interferometer (off-axis und phase-stepping mit Piezo-Aktuatoren); die Phasenrekonstruktion erfolgt über iterative Algorithmen (Zernike-Dekomposition), implementiert in Python und C++. Das System steuere ich in Echtzeit über LabVIEW/FPGA und nutze es täglich zur Aberrationsmessung. Zwei Publikationen (ACS Photonics, Biomedical Optics Express) dokumentieren die Ergebnisse. Parallel dazu habe ich ein Multiphotonen-Mikroskop komplett aufgebaut: Femtosekundenlaser-Integration, Galvo-Scanner-Ansteuerung, PMT-Signalerfassung und ZEMAX-Toleranzanalyse der Mikroskopobjektive.

Am Leibniz-IPHT in Jena habe ich zuvor einen 1064 nm Raman-Spektrometer-Prototyp von der optischen Auslegung über die Kalibrierung bis zur Publikation (Applied Sciences) durchgeführt. Dies entspricht einem vollständigen Prototypenzyklus, wie er auch bei SmarAct für den Übergang von Entwicklung zur Serie gefragt ist. Aus meiner früheren Tätigkeit als Machine Vision Engineer bringe ich zusätzlich Erfahrung in der Entwicklung optischer Inspektionssysteme und deren Überführung in die Serienproduktion mit.

Mein frühestmöglicher Eintrittstermin ist der 1. Juli 2026. Ich bin uneingeschränkt arbeitsberechtigt und freue mich auf ein persönliches Gespräch.

Mit freundlichen Grüßen,

Juan David Muñoz-Bolaños

Doktorand, Medizinische Universität Innsbruck

Juan David Muñoz-Bolaños

Optical System Engineer

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Summary

Physicist and optical engineer with deep expertise in **laser interferometry, wavefront sensing, and precision optical metrology**. Proven track record in end-to-end prototype development, from optical concept and simulation (ZEMAX, Python) through hardware-software integration (Python, C++, LabVIEW) to validated, publication-grade systems. Strong analytical skills and experience transitioning R&D concepts into robust, reproducible measurement setups. Author of three peer-reviewed publications in applied optics and photonics.

Professional Experience

Jan 2022 to Present **Medical University of Innsbruck** Innsbruck, Austria
PhD Researcher, Laser Interferometry & Adaptive Optics

Interferometric Wavefront Sensing: Designed and built a Michelson-type interferometric wavefront sensor (off-axis and phase-stepping with piezo actuators) for adaptive optics microscopy; implemented iterative phase retrieval algorithms and a C++/Python/LabVIEW control stack for scientific cameras and deformable mirrors.

Prototype Development & Validation: Led the end-to-end assembly and commissioning of a custom nonlinear (multiphoton) microscope, including fs-laser integration, galvo-scanner control, and PMT signal acquisition, achieving diffraction-limited resolution.

Physical Modelling & Simulation: Performed systematic ZEMAX tolerance analysis and wave-field simulations (Zernike decomposition) to characterize aberration sensitivity of custom microscope objectives.

Technical Documentation: Published three peer-reviewed articles in ACS Photonics, Biomedical Optics Express, and Applied Sciences, demonstrating whitepaper-level scientific writing.

Jun 2020 to Dec 2021 **Leibniz Institute of Photonic Technology (IPHT)** Jena, Germany
R&D Research Assistant, Photonics & Spectrometer Prototyping

Spectrometer Prototyping: Designed, built, and calibrated a dispersive 1064 nm fiber-probe Raman spectrometer for tissue characterization; published results in Applied Sciences.

Data Acquisition & Analysis: Developed 'SpectraMap' (custom Python package) for supervised ML classification of hyperspectral Raman images, optimizing measurement throughput and evaluation.

Mar 2018 to Mar 2019 **INTECOL S.A.S.** Medellín, Colombia
Junior Machine Vision System Engineer

Optical System Design: Engineered custom illumination (dark-field, bright-field, coaxial) and Halcon pipelines for bottle labeling lines (liquid levels, cap sealing); implemented Cognex VisionPro for automatic part recognition in automotive lines.

R&D-to-Production: Transitioned prototype vision setups to production-ready deployments; performed Gage R&R assessments and feasibility studies for client sign-off.

Technical Competencies

Laser Interferometry & Metrology: Wavefront sensing, phase retrieval, Michelson interferometry, adaptive optics, precision alignment of laser-based measurement systems.

Prototyping & Optical Design: End-to-end system build from concept to validated prototype. ZEMAX/OpticStudio modelling, tolerance analysis, opto-mechanical integration, component qualification (Thorlabs, Edmund Optics, Hamamatsu).

Data Acquisition & Simulation: Python, C++, LabVIEW/FPGA, MATLAB for real-time data capture, signal processing, physical modelling, and instrument control. Experience with JAX (GPU computing).

Technical Documentation: Three peer-reviewed publications (ACS Photonics, Biomedical Optics Express, Applied Sciences). Experienced in writing application notes, technical specifications, and whitepaper-style documents.

Industrial Optics: Halcon & Cognex VisionPro pipelines, custom illumination design, camera integration, Gage R&R, system commissioning on production lines.

Education

Jan 2022 to Present	Medical University of Innsbruck <i>Ph.D. in Adaptive Optics & Microscopy</i>	Austria
2019 to 2021	Friedrich Schiller University Jena <i>M.Sc. in Photonics</i>	Germany
2012 to 2018	University of Cauca <i>B.Sc. in Physics Engineering</i>	Colombia

Publications & awards

Awards: ASML & ZEISS Summer Schools participant; COLFUTURO Grant 2026 awardee.

Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).

Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).

Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

Languages

English (Professional) · German (B2, intermediate/ working proficiency and ongoing) · Spanish (Native)

Soft skills

Collaboration & Communication: Communicated complex technical results through peer-reviewed publications, conference presentations (SPIE Photonics West), and international seminars within cross-functional R&D teams.

Problem Solving: Tackled complex, cross-disciplinary challenges resulting in three peer-reviewed publications and a conference proceeding presented at SPIE Photonics West, San Francisco.

Adaptability: Rapidly integrated into three distinct research environments across three countries (University of Cauca, Leibniz-IPHT Jena, Medical University of Innsbruck), each with different workflows, toolchains, and scientific cultures.

Hobbies

Mountain Biking, Dancing Bachata, Learning Harmonica, Learning German.

References

Prof. Monika Ritsch-Marte

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Spectroscopy Group Leader
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Leibniz Institute of Photonic Technology